Problem Pack: Pythagorean Theorem Applications, Mathematics 8 - Unit 2 Review

Question 17

Problem: 17. A wheelchair ramp rises 1.5 m over a horizontal run of 3.6 m. Find the length of the ramp surface. Round to two decimal places where needed.

Student Work:

c^2 = 1.5^2 + 3.6^2 = 2.25 + 12.96 = 15.21 c = √15.21 = 3.90

Feedback

Question 18

Problem: 18. A triangular garden has two perpendicular sides measuring 9 m and 12 m. Find the third side, then find the total length of fencing needed to surround the garden.

Student Work:

c^2 = 9^2 + 12^2 = 81 + 144 = 225 c = 15 = 15

Feedback

In question 18, you went wrong when calculating the total length of fencing by only adding the third side instead of all three sides together. You forgot that fencing has to go all the way around the outside of the garden, covering every side. Try adding the lengths of all three sides of the garden to find the total distance around it.

Question 19

Problem: 19. Two boats leave the same marina at the same time. One travels north at 5 km/h and the other travels east at 12 km/h. How far apart are the boats after 2 hours? Show how the situation forms a right triangle.

Student Work:

N: 5 = 10 E: 12 = 24 c^2 = 10^2 + 24^2 = 100 + 576 = 676 c = √676 = 26

Feedback

Question 20

Problem: 20. A two-part support is made from two right triangles. In the first triangle, the legs are 7 cm and 24 cm. Its hypotenuse becomes one leg of a second right triangle whose other leg is 60 cm. Find the final length x. Show both uses of the Pythagorean theorem.

Student Work:

1: c^2 = 7^2 + 24^2 = 49 + 576 = 625 = 25 2: x^2 = 25^2 + 60^2 = 625 + 3600 = 4225 √4225 = 65

Feedback